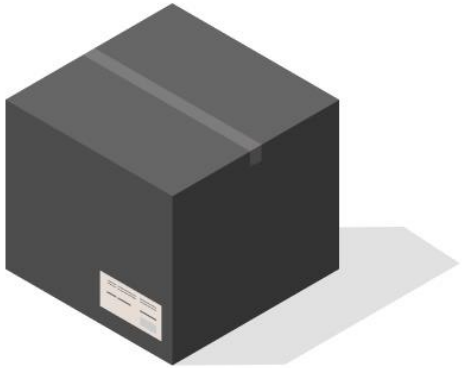
# Natural Language Technology

Week 5 – Structured Text Analysis

# Recap Week 3

- Representing Text for Natural Language Processing
- We want to do more than represent the characters for easy edit and save
- Various approaches
  - One Hot Encodings
  - Bag of Words
  - TF-IDF
  - Word Embeddings
  - Sentence Embeddings
- These representations make it easy to perform tasks like similarity measurement
  - This is surprisingly important, and we will come back to it later

# Black Box vs White Box



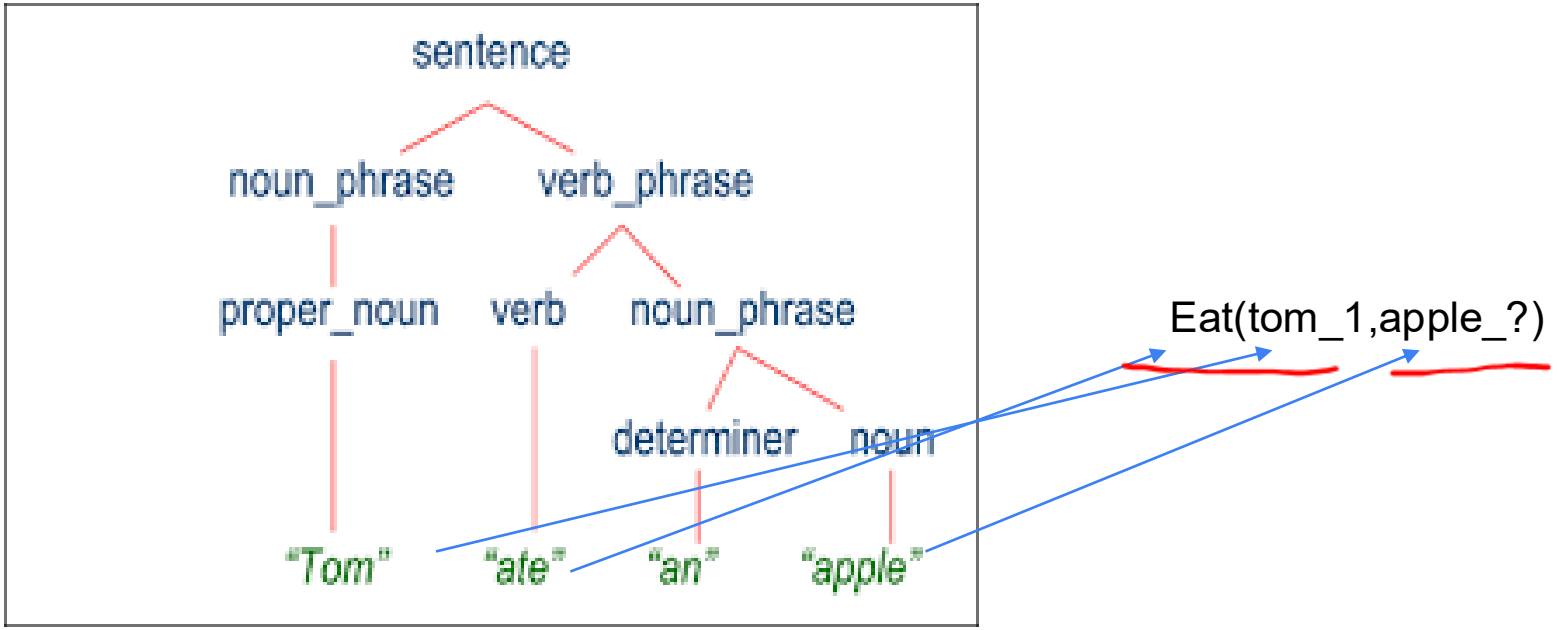
Black box - we do not know anything



White box - we know everything

The concept of black-box versus white-box is an important concept in software testing, but we can take the analogy and also apply it to our understanding of information. How can say a text classifier be white box versus black box?

# White Box: sometimes we need the specifics:




A white-box approach to text processing suggests that we make clear that we understand the bits that go into a text or document. This can help with making tasks transparent.

# Today

- Getting to the specifics
- Parsing
- POS Taggers
- Entity Extraction
- Using Spacy

# Today

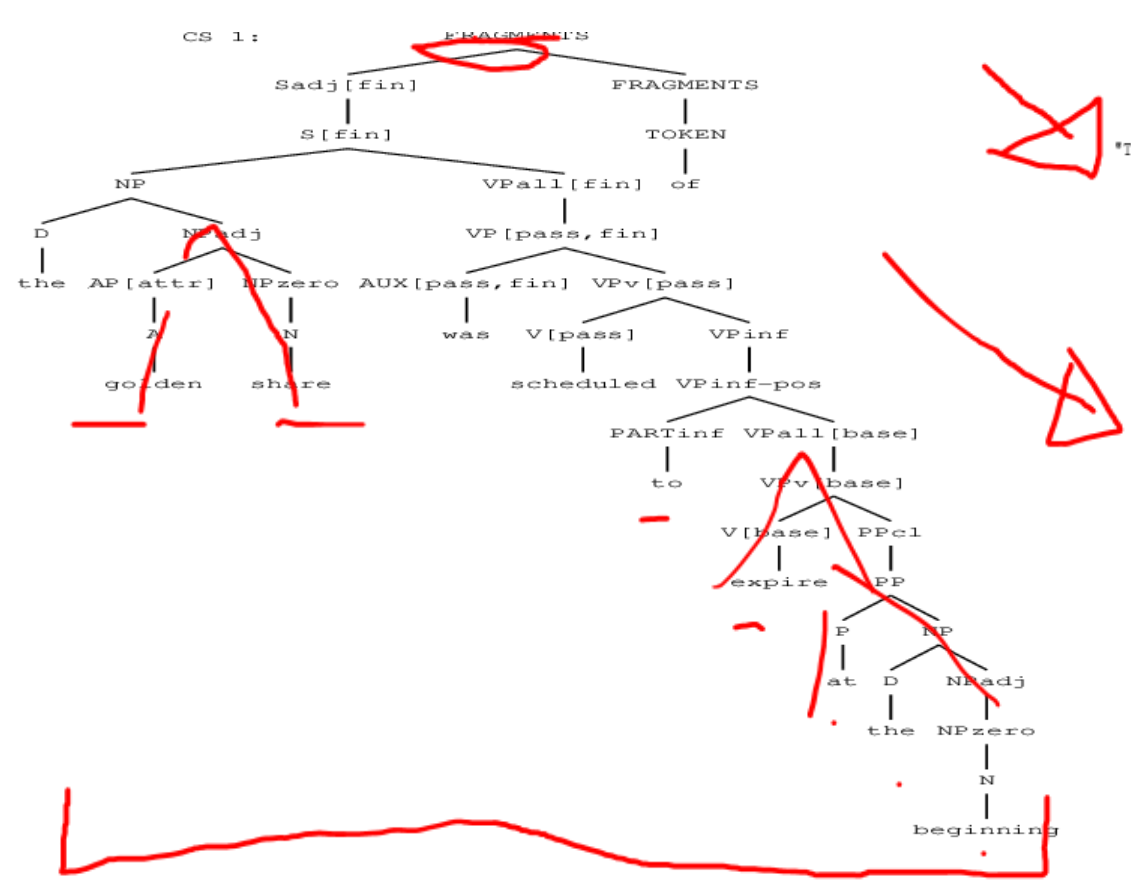
- Getting to the specifics
- Parsing
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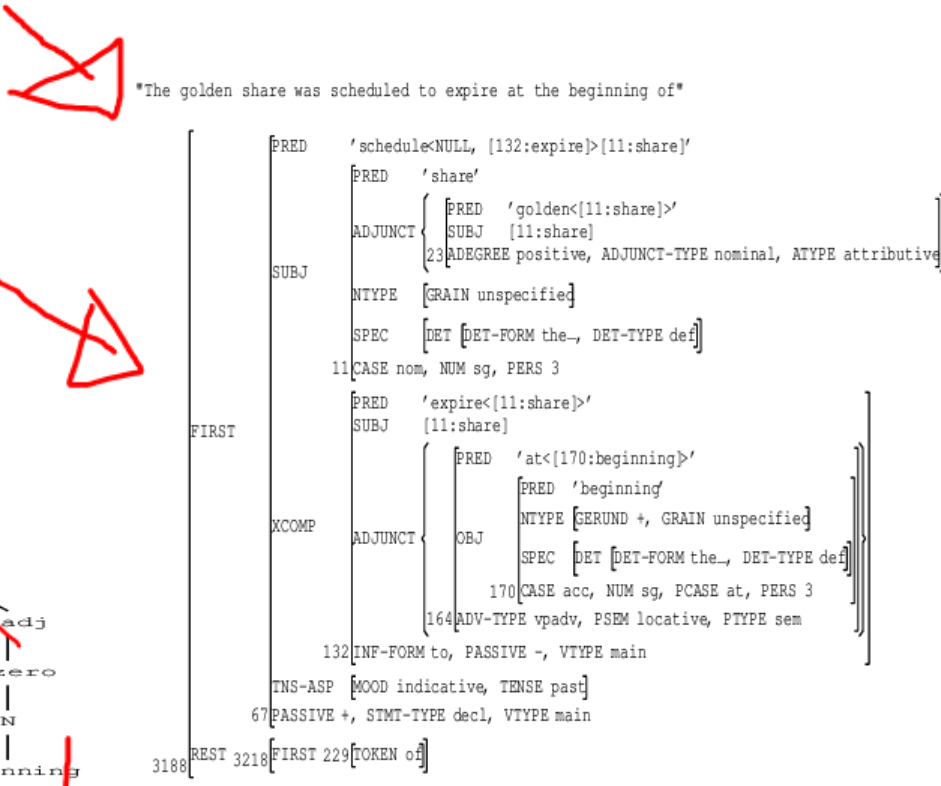
Remember: We are doing just enough on these methods to understand their role, and to best understand the relationships between various concepts in NLT and related areas such as formal parsing.

# Parsing

# Parsing and Analysis

- **Parsing** is classically thought of dividing a sentence or other document into its constituent pieces, often through the use of well-defined rules, and with clear expectations in terms of what individual elements can be.
  - Roots in linguistics theory
  - Many different approaches in Computational Linguistics and NLP
  - Specific forms are very important in programming
  - Often assumed to produce a complete well-structured output




**From: Parsing the Wall Street Journal using a Lexical-Functional Grammar and Discriminative Estimation Techniques**

Left: Traditional Parse Tree

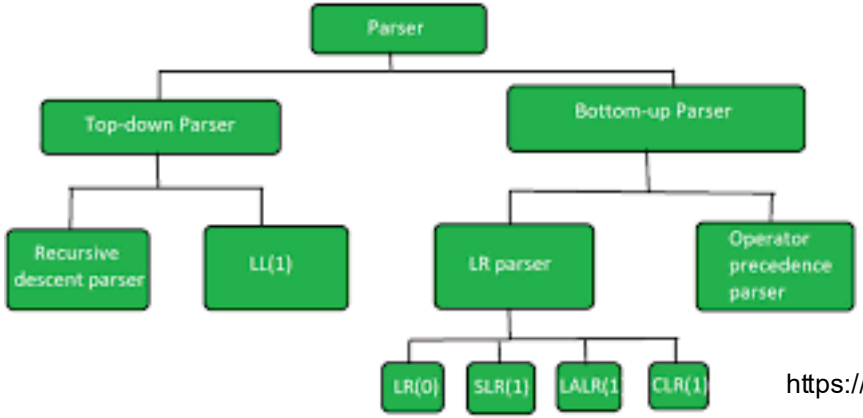
Right: Dependency Structure

# Parsing and Analysis

- **Parsing** is classically thought of dividing a sentence or other document into its constituent pieces, often through the use of well-defined rules, and with clear expectations in terms of what individual elements can be.
  - Roots in linguistics theory
  - Many different approaches in Computational Linguistics and NLP
  - Specific forms are very important in programming
  - Often assumed to produce a complete well structured output
- **Analysis** is the lazy but often more effective cousin of parsing that focusing on extracting the relevant types of information from text for a specific task. Sometimes it is based on a parse structure, but often it can be achieved using machine learning methods without a true parse.

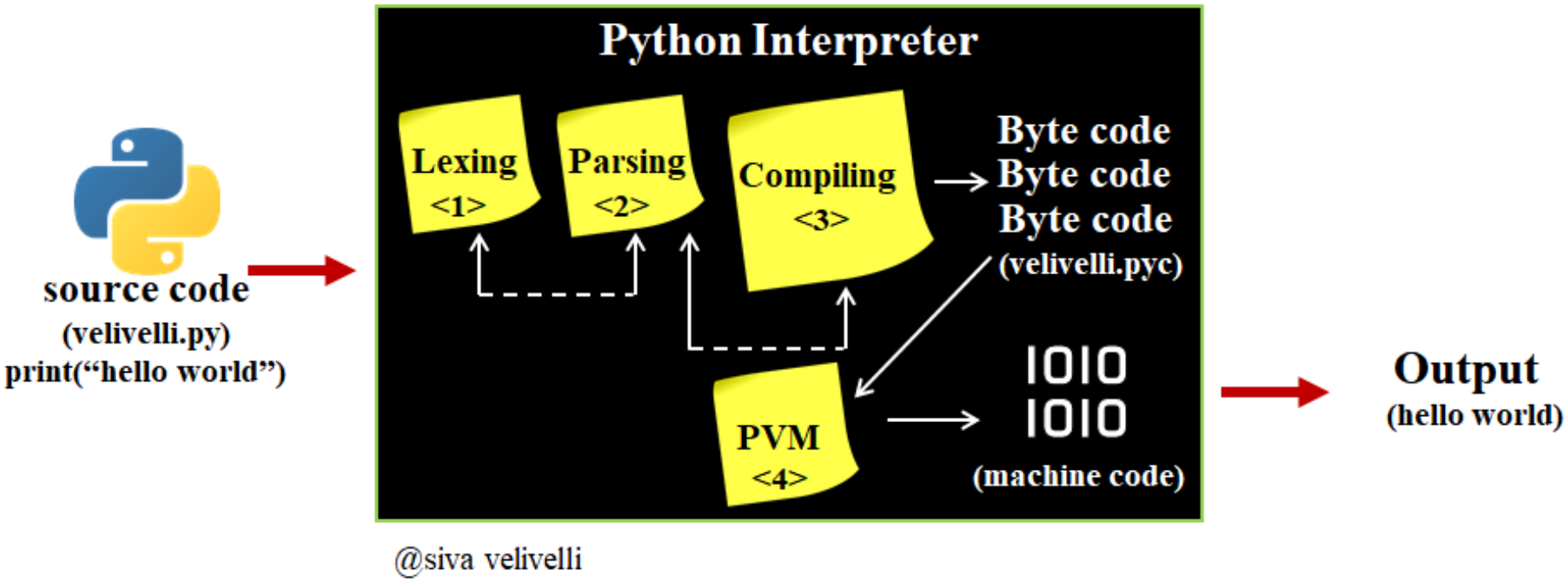
# Comp Sci Parser Types

- Since parsers are so fundamental to much of computer science, there are many different types of types of parsers! For example, within the strict computer science domain, below is a common breakdown of parsers. This is not in our scope, and I don't want you to try to remember or investigate it. Just remember that it exists, and it is distinct from the parsers we are looking at.



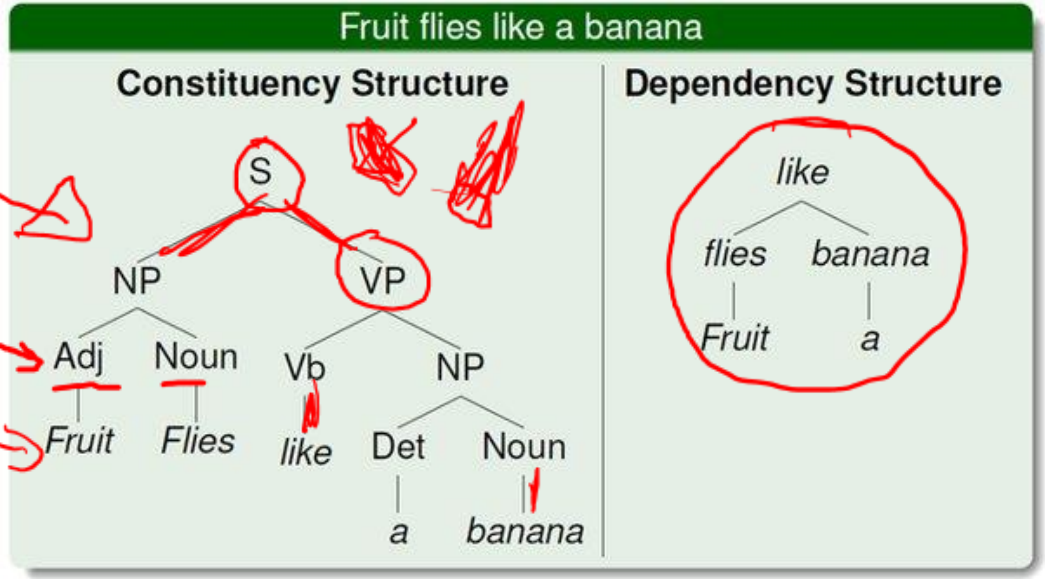

https://www.geeksforgeeks.org/types-of-parsers-in-compiler-design/

# Python: Example of ‘Formal Language’ Processing




Source: https://www.linkedin.com/pulse/how-does-python-interpreter-work-siva-velivelli/

# **Natural Language** Parser Types

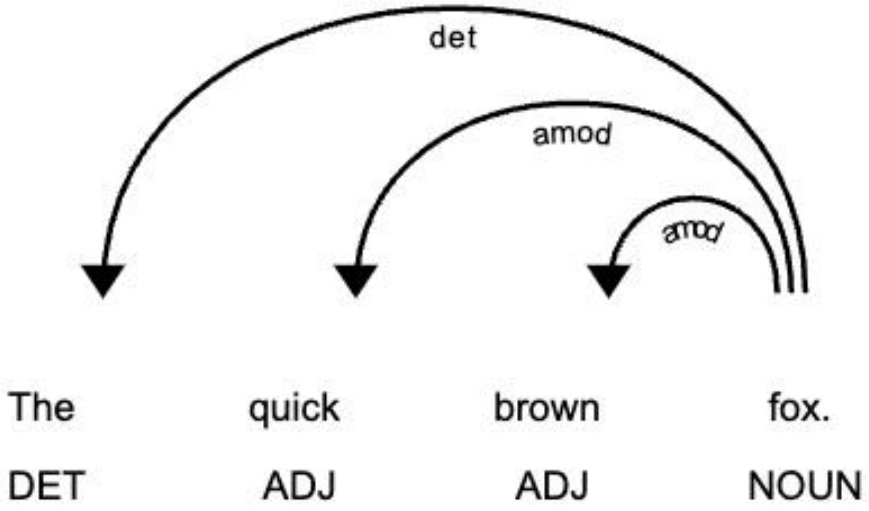



**Constituency Parsers** try to break a sentence down in terms of its Parts-of-Speech. This is the linguistic common sense grammatical categories like 'noun' 'verb' etc. Rules are used to define what parts of speech can be rolled up into more high-level parts of speech.

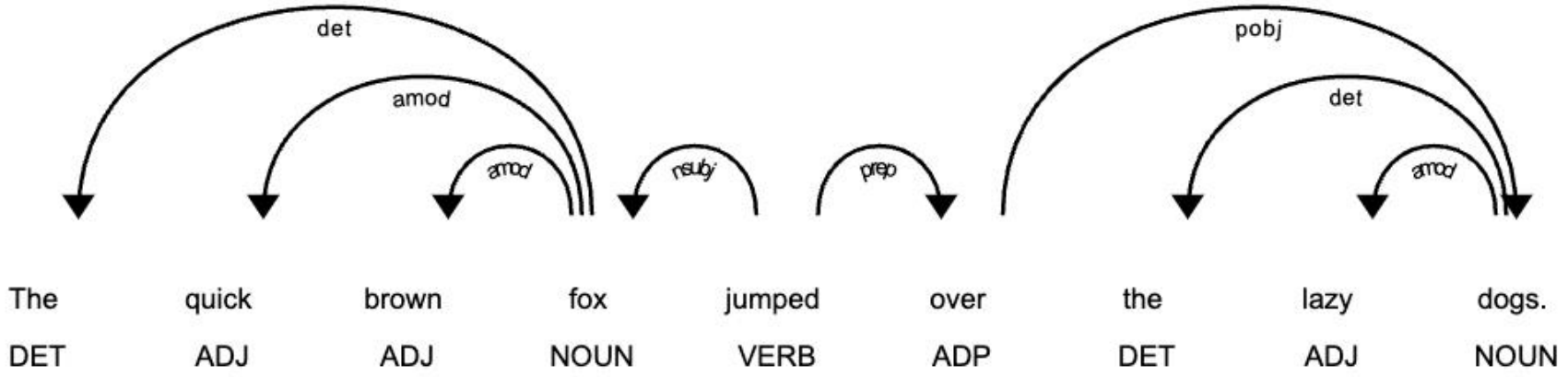
**Dependency Parsing** focuses more on the relationships between words according to grammatical rules. While it is still thought of as being grammar focused, it is closer to a meaning focused interpretation

# Dependency Parsing

SPACY



# Dependency Parsing




**Parse structures unveil the function of words within a document by establishing the specific roles between elements in the text.**

# Parsing Mechanism

- There are many different ‘parser types’ that can be used to build a parse tree for a given document. Given its relative simplicity and transparency, let’s take the Probabilistic Context Free Grammar grammar as an example:

Start Symbol

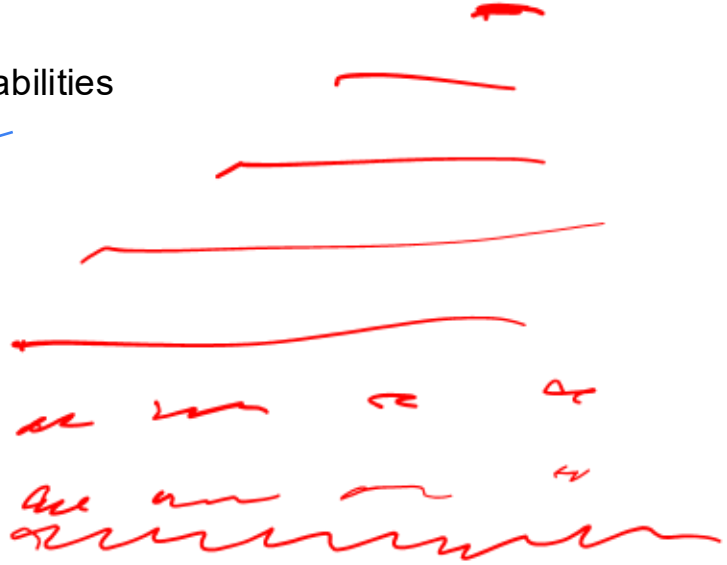
Probabilities

| | |
|---|---|
| $S \rightarrow NP\ VP$ | .8 |
| $S \rightarrow VP$ | .2 |
| $NP \rightarrow D\ N$ | .4 |
| $NP \rightarrow NP\ PP$ | .4 |
| $NP \rightarrow PN$ | .2 |
| $VP \rightarrow V\ NP$ | .7 |
| $VP \rightarrow VP\ PP$ | .3 |
| $PP \rightarrow P\ NP$ | 1 |

Rewrite Rules

| | |
|---|---|
| $D \rightarrow the$ | .8 |
| $D \rightarrow a$ | .2 |
| $N \rightarrow flight$ | 1 |
| $PN \rightarrow john$ | .9 |
| $PN \rightarrow schiphol$ | .1 |
| $V \rightarrow booked$ | 1 |
| $P \rightarrow from$ | 1 |

Terminal Rules



# Parsing Mechanism

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| $PP \rightarrow P\ NP$ | 1 |

Rewrite Rules

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| $P \rightarrow from$ | 1 |

Terminal Rules

How do we come up with the rules?

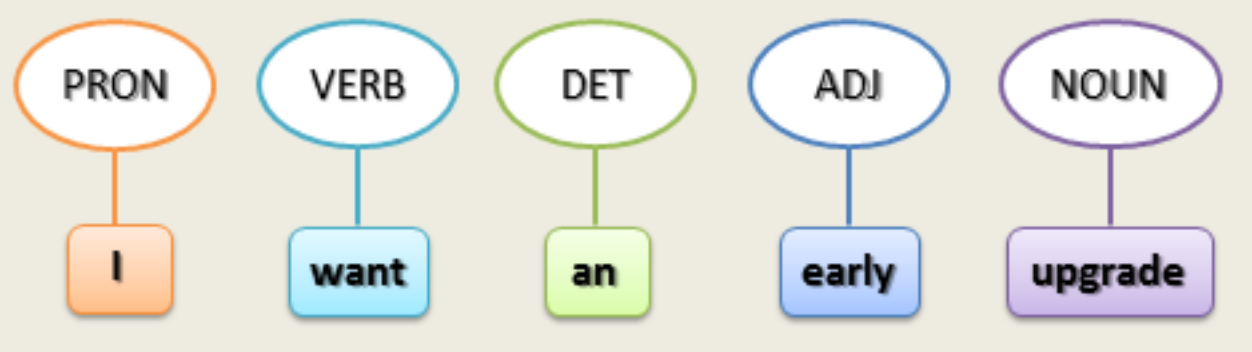
- Hand crafted?
- Learned from the data

# Part of Speech Tagging

# Parts of Speech

- A part of speech is a category of word that demonstrates certain grammatical properties.
- There are a relatively small number of Parts of Speech (POS) types / categories in English:
  - Noun (“dog”)
  - Verb (“eat”)
  - Pronoun (“her”)
  - Adjective (“quick”)
  - Adverb (“quickly”)
  - Preposition (“over”)
  - Conjunction (“and”)
  - Interjection (“wow”)
  - Article (“the”)
- Parts of speech are useful features for differentiating between words and assisting analysis.

# POS Tagged Sentences



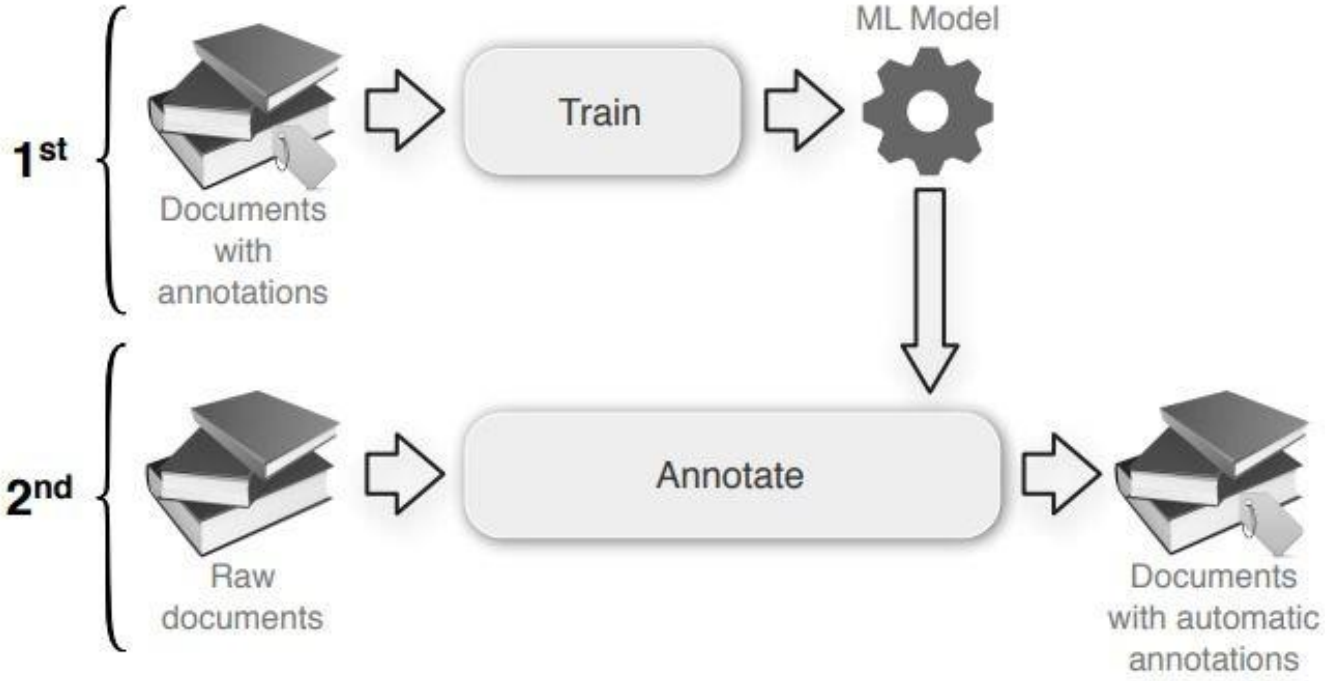

https://thinkinfi.com/extract-custom-keywords-using-nltk-pos-tagger-in-python/

# Part of Speech Tagging - Process

- Overall Process:
  - Collect & Label
    - Collect a large corpus of text
    - Get humans to label the text according to some annotation schema
  - Train Model
    - Use some supervised learning process to train a statistical or more generally a machine learning model based on the data
      - Statistical models use probabilities
      - Other machine learning models will use other mechanisms which correlate with statistics but don't always perfectly match
  - Use Model
    - Apply the trained model to new unseen data

Note that this process is very typical to many other related challenges in working with text data. While we introduce it for POS, it is not specific to POS tagging

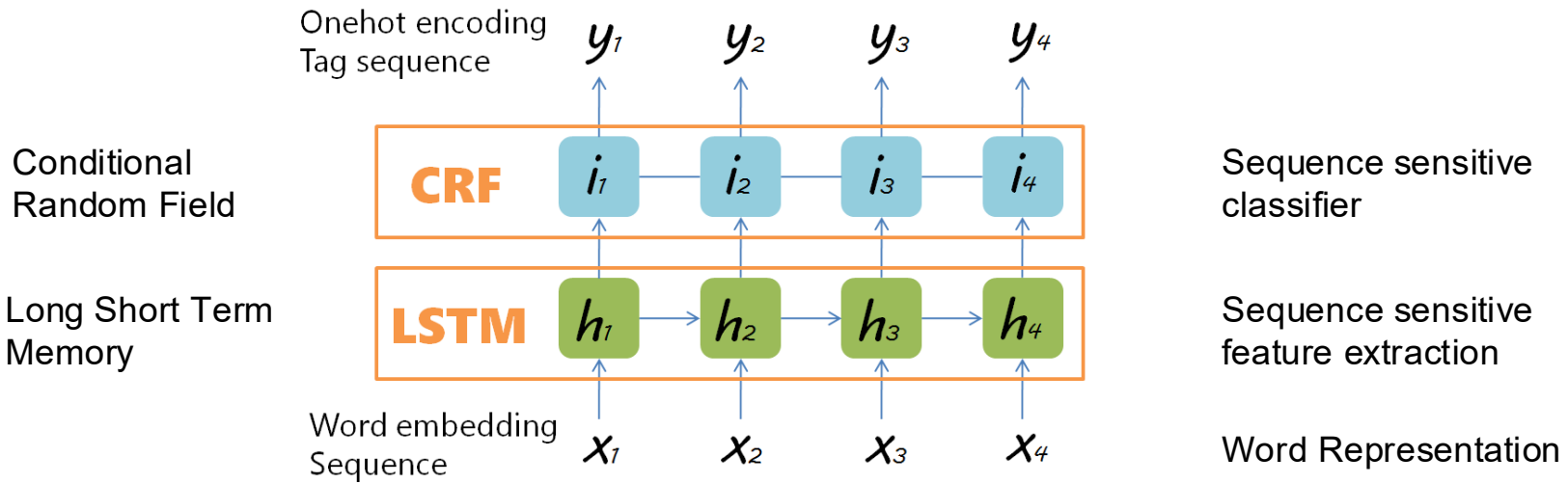
# The Supervised Process for Text Systems




https://www.turing.com/kb/a-comprehensive-guide-to-named-entity-recognition

# Part of Speech Tagging Models

- Use of any mechanisms – usually now machine learning based – to tag a sentence with associated parts of speech




The architecture of LSTM+CRF model

https://github.com/wantinghuang/tensorflow-lstmcrf-postagger

# Named Entity Recognition

# Named Entity Recognition

- Named Entity Recognition (NER) is the process of identifying the key entities that are named in a document.
  - We want to take the entities and classify them into pre-defined types
- NER unlike parsing focuses on just the People, Places, Organizations etc that are to be found in a document.
- Usage: Information Retrieval and also non-vector based document processing tasks

# Examples of Named Entities



The Liffey

Bank of Ireland

Bank of Ireland

# Examples of Named Entities



The Liffey

Bank of Ireland

Bank of Ireland

Banc Ceannais na hÉireann
Central Bank of Ireland
Eurosystem



Central Bank of Ireland

# Examples of Named Entities



The Liffey

RED RIVER BANK

Banc Ceannais na hÉireann
Central Bank of Ireland
Eurosystem



Central Bank of Ireland

# More Examples of Named Entities



Buck Lane – Real Person

Paris Hilton



# More Examples of Named Entities



Buck Lane – Real Person

Paris Hilton





Paris Hilton?

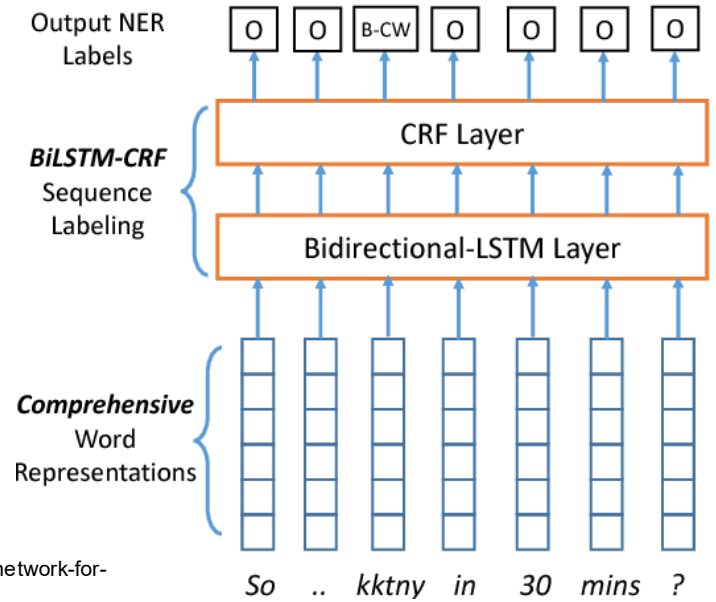
# Challenges in NER

- Ambiguity
  - Often context fixes it
- New terms
  - A neologism is a newly coined word or expression
  - Business or organization names frequently come into new regular use
- Named entities are an ‘open class”

# NER Training and Use

- The training and use of NER systems is very like the case for POS taggers, except that the open class nature of named entities makes the task more challenging

Note that the typical architecture is very similar to the POS case. Word representations, LSTM based feature extractors, and a CRF layer to make the predictions




https://confusedcoders.com/data-science/deep-learning/how-to-build-deep-neural-network-for-custom-ner-with-keras

# NER Tags

Depending on the dataset used to train the Named Entity Recognition software, you can have a different set of NER tags. Below is the list of tags built into the Spacy tool's NER pipeline for English. Note that other languages will have similar but different sets.

| | |
|---|---|
| NER ⓘ | `CARDINAL`, `DATE`, `EVENT`, `FAC`, `GPE`, `LANGUAGE`, `LAW`, `LOC`, `MONEY`, `NORP`, `ORDINAL`, `ORG`, `PERCENT`, `PERSON`, `PRODUCT`, `QUANTITY`, `TIME`, `WORK_OF_ART` |

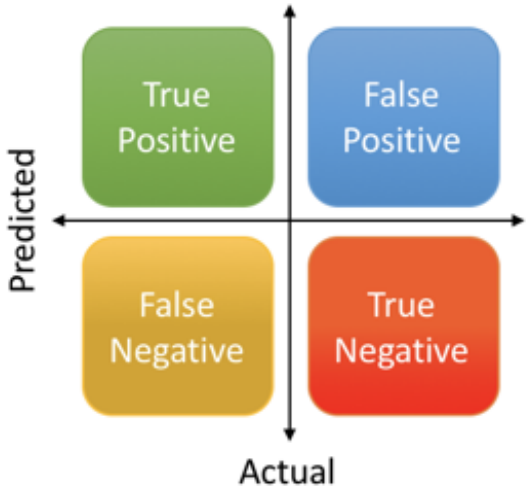
# Measuring Correctness

For both software users and model builders, it is important to understand that all these systems need to be trained with but evaluated against some measure of of correctness. For simple binary classification tasks, we often use accuracy (see below) but it can be dangerous if our classes are not balanced.

$$\textbf{Precision} = \frac{\text{True Positive}}{\text{Actual Results}} \quad \text{or} \quad \frac{\text{True Positive}}{\text{True Positive + False Positive}}$$

$$\textbf{Recall} = \frac{\text{True Positive}}{\text{Predicted Results}} \quad \text{or} \quad \frac{\text{True Positive}}{\text{True Positive + False Negative}}$$

$$\textbf{Accuracy} = \frac{\text{True Positive + True Negative}}{\text{Total}}$$




https://www.linkedin.com/pulse/precision-recall-binary-classifier-model-evaluation-prakruthi-b-gowda/

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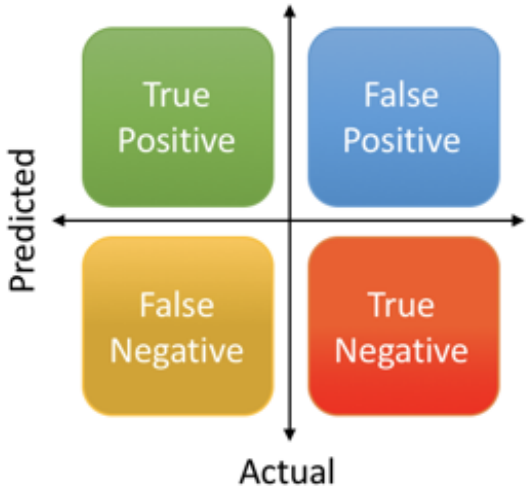
$$\textbf{Precision} = \frac{\text{True Positive}}{\text{Actual Results}} \quad \text{or} \quad \frac{\text{True Positive}}{\text{True Positive + False Positive}}$$

How many of the things we said are positive, are positive?

$$\textbf{Recall} = \frac{\text{True Positive}}{\text{Predicted Results}} \quad \text{or} \quad \frac{\text{True Positive}}{\text{True Positive + False Negative}}$$

How many of the things that should have been positive, are positive?

$$\textbf{Accuracy} = \frac{\text{True Positive + True Negative}}{\text{Total}}$$




https://www.linkedin.com/pulse/precision-recall-binary-classifier-model-evaluation-prakruthi-b-gowda/

# Measuring Correctness with the F1 score

The F1 score is the so-called harmonic mean of precision and recall. A harmonic mean is different to a normal mean. For our purposes it is important to note that a harmonic mean is more conservative or pessimistic than a regular mean. It will give a value that is somewhere between the min of two values and the actual mean.

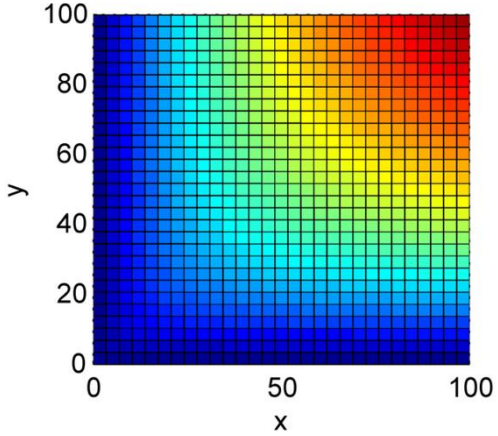
$$\text{F1 Score} = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}}$$

$$= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

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$$\text{F1 Score} = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$




Visualization of harmonic mean (F1 score). What would you expect this plot to look like for a normal / arithmetic mean?

# Important Side Note

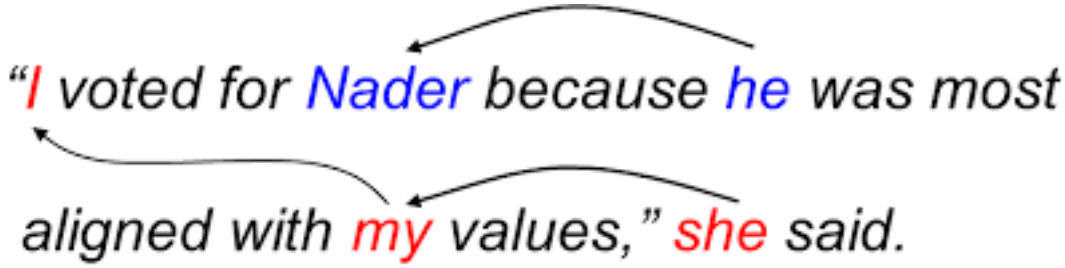
- For many Natural Language Technology tasks, it is easy to build a basic system that maybe covers 80% of cases, it is tough but doable to cover 90-95% of cases, but it is the examples you miss that will often define the perceived quality of a solution

- Similarly, it is easy to build a mobile app, it is hard to build a good mobile app

# Coreference Resolution

# What is coreference resolution

- Related terms: Reference resolution, anaphora resolution
- This is the process of determining the expressions that refer to the same entities
  - Pronoun resolution
  - Even relating two instances of John to each other




https://nlp.stanford.edu/projects/coref.shtml

# Coreference Resolution Support

- While co-ref resolution has been studied widely in research, in practice it is still a little flaky in tools and APIs.
- SpaCy has added experimental support for this. We will look at this in the lab notebooks.

# Processing Dialogue

# Turns and Dialogue Acts

- Turns and Dialogue Acts are important structural elements in conversation
- A Turn is a point in a dialogue where one person holds the floor while the other person listens. Turn taking is an important part of achieving fluid conversation, and turn-taking as an academic exercise is an important task in understanding the dynamics of a conversation
- A dialogue act is a pragmatic label we can apply to a chunk of a conversation or dialogue to note where something important to the dialogue flow occurring. More formally, but less clear, “A dialog act is a **specific component of speech that labels a piece of dialog according to its category** of meaning. ”

# Dialogue Turns

I'm starving, have you put the dinner on?

No, I haven't had a chance, the dog barfed in his bed, and I just spent the last 20 minutes cleaning up that mess. When are you leaving?

I've about 20 minutes left, but I'll leave then

This sequence is ideal. In the real world there will be interruptions and overlaps which blur the edges in terms of what is a turn and what isn't

# Dialogue Acts (Speech Acts)

An action oriented interpretation of what a speaker intended to achive with a contribution.

In a simplified world, there can be a tight link between grammatical sentence types and a dialogue act:

Statement = Inform
Question = Ask-if
Declarative = Request

| Speech act | Description | Example | Percent (%) of corpus |
|---|---|---|---|
| Greeting | The opening greeting of a dialogue | Hello. | 9.43 |
| Closing | The closing greeting of a dialogue | Good-bye. | 8.79 |
| Opening | Sentences for opening a goal-oriented dialogue | May I help you? | 0.02 |
| Ask-ref | WH-questions | Where is the place? | 22.78 |
| Ask-if | YN-questions | Can I change the time? | 2.76 |
| Response | Responses to questions or requesting actions | Yes, you can. | 37.98 |
| Request | Declarative sentences for requesting actions | Set the alarm. | 14.24 |
| Ask-confirm | Questions confirming the previous actions | Saturday, right? | 0.03 |
| Confirm | Reponses to ask-confirm | Right. | 0.03 |
| Inform | Declarative sentences giving some information | It was canceled. | 2.07 |
| Accept | Agreement | I know. | 1.88 |

Baseline Dialogue Act Set

https://www.researchgate.net/figure/Speech-acts-and-their-meanings_tbl2_285847014

# Indirect Dialogue Acts

- Not all speech acts are so direct. Indirect speech acts can be used to achieve goals in less obvious ways if we focus only on the content of the sentence, but mure more obvious when we understand conversational norms and the context of a conversation.
- For example, what might the true meaning of these utterances be:
  - Its gone very cold in here
  - I could do with a cup of tea
  - It would be a lot easier to make progress if we had more information

| Speaker | Text | Utterance (Units of Speech) | Turn (Speaker's Time Slot) | Dialogue Act (Function/Intent) |
|---|---|---|---|---|
| Alice | "Hey, Bob! How are you doing today?" | U1 | T1 | Greeting (Initiation) |
| | "Did you manage to find that book?" | U2 | T1 | Question (Information Request) |
| Bob | "Hi, Alice." | U3 | T2 | Greeting (Response) |
| | "I'm great, thanks! And yes, I did." | U4 | T2 | Statement (State Report) |
| | "I picked it up from the library this morning." | U5 | T2 | Inform (Additional Context) |
| Alice | "Fantastic!" | U6 | T3 | Acknowledge (Positive Feedback) |
| | "Can I borrow it on Thursday?" | U7 | T3 | Request (Action Elicitation) |
| Bob | "Sure, no problem." | U8 | T4 | Accept (Commitment) |
| Alice | "Thanks!" | U9 | T5 | Thank (Expression of Gratitude) |

# Processing Dialogue

# Dialogue Act Tagging

- Just as we can build POS taggers to analyse the part of speech contribution of a word to a sentence, we can build dialogue act taggers to try to estimate the contribution of one or more contributions to the overall dialogue
- This is by no means as reliable as POS tagging

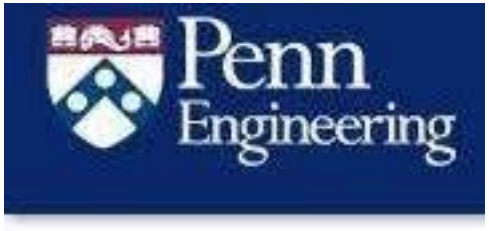
# A More Basic Approach: Intent Recognition

- Instead of having to figure out a full dialogue act / dialogue tag representation for a 3rd party dialogue, in conversational systems we usually just need to figure out the ‘intent’ of the user.
- There are often overlaps between dialogue act ideas and how we talk about intents, but intent recognition is simplified considerably by the constraints of the interaction
    - What is the task?
    - What is it likely the user will ask about?
    - Etc etc

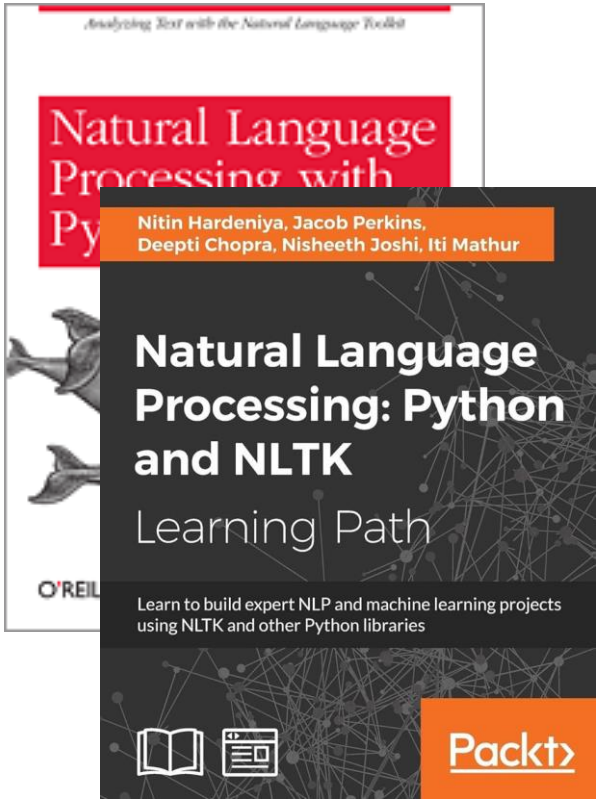
# Tool Support

# The Natural Language Toolkit




Computer &
Information
Science



# spaCy




Like NLTK, it is a standalone tool that provides support for multiple languages for key natural language processing tasks.

Unlike NLTK, it was designed from the ground up to be a tool fit for use in a commercial environment



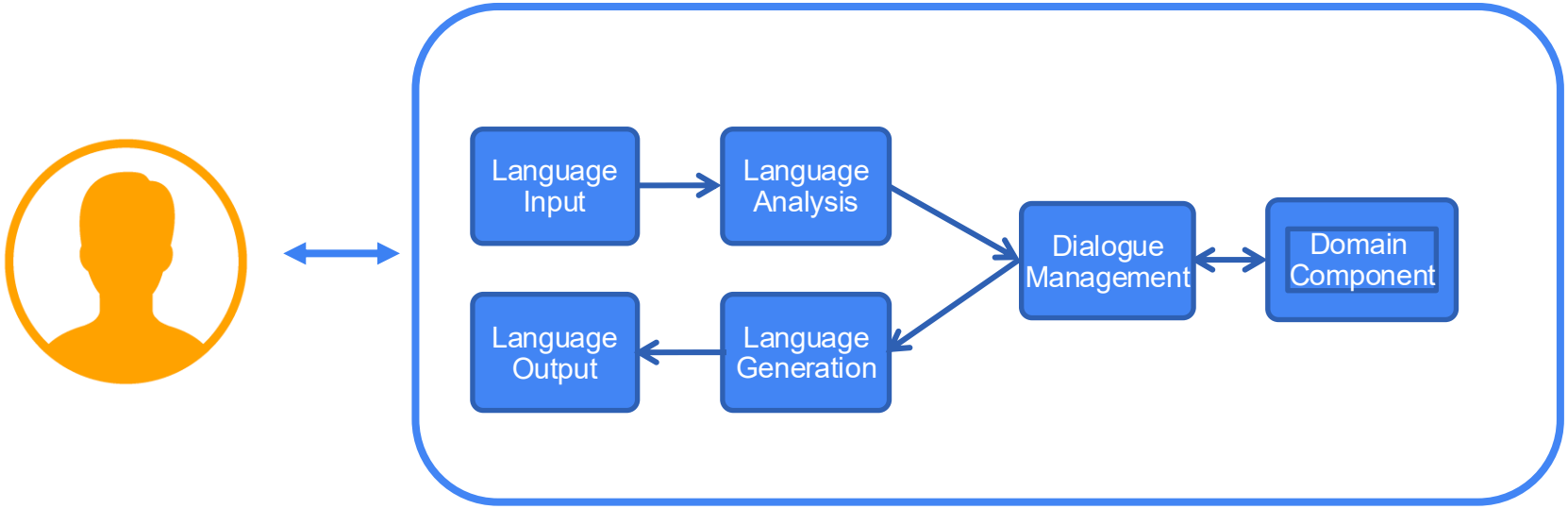
Google
Natural Language
API



IBM Watson

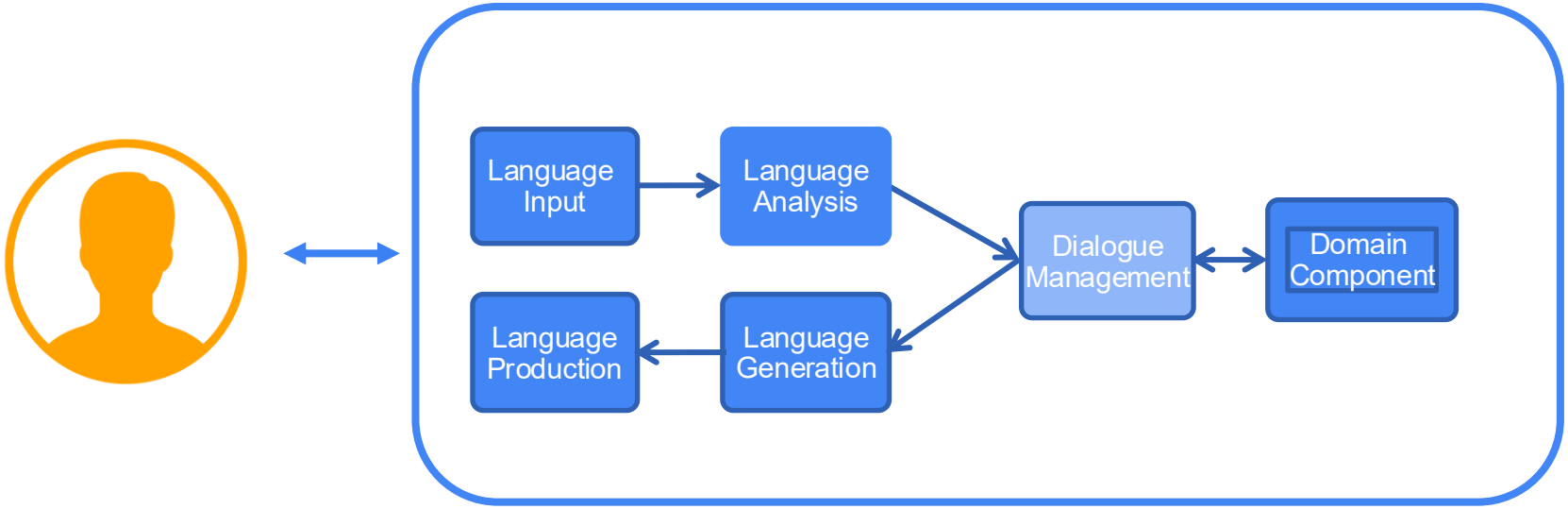
# Dialogue Management

# Conversational Architectures

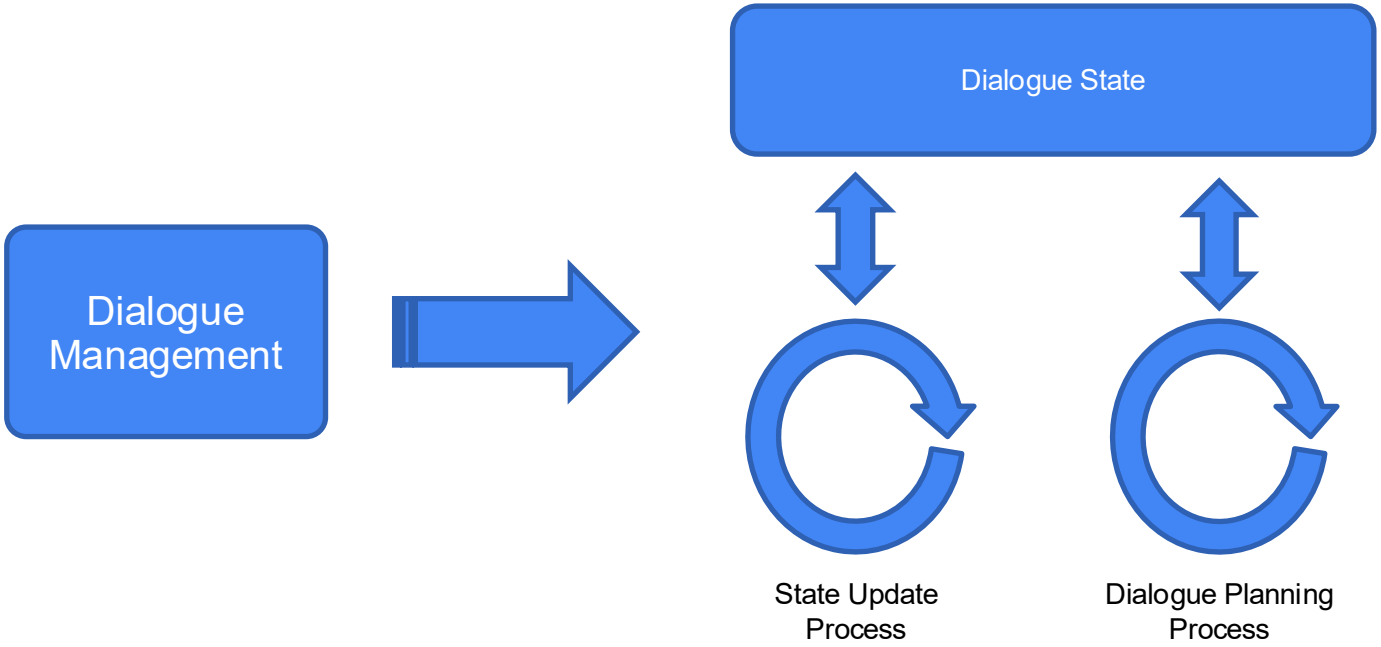



*Architectures of natural language technology components used to enable conversational applications*

# Dialogue Management



# State Management and Planning

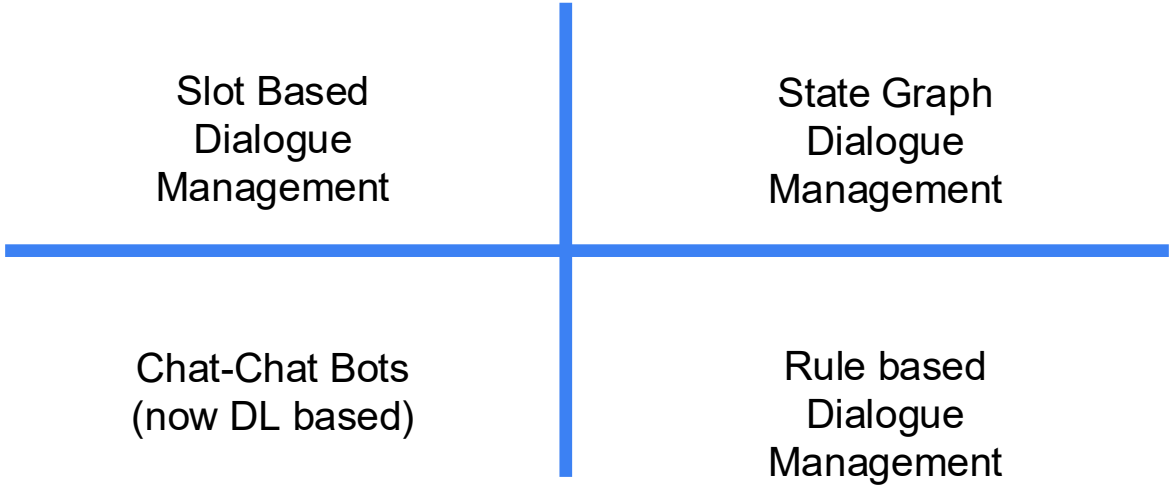


# Dialogue Management Processes

- Dialogue State Management: Updating the dialogue state in response to a user utterance.
- Dialogue Planning: Deciding on the action to be taken (dialogue act or other act) at the current time given the dialogue state (including the user's last utterance)

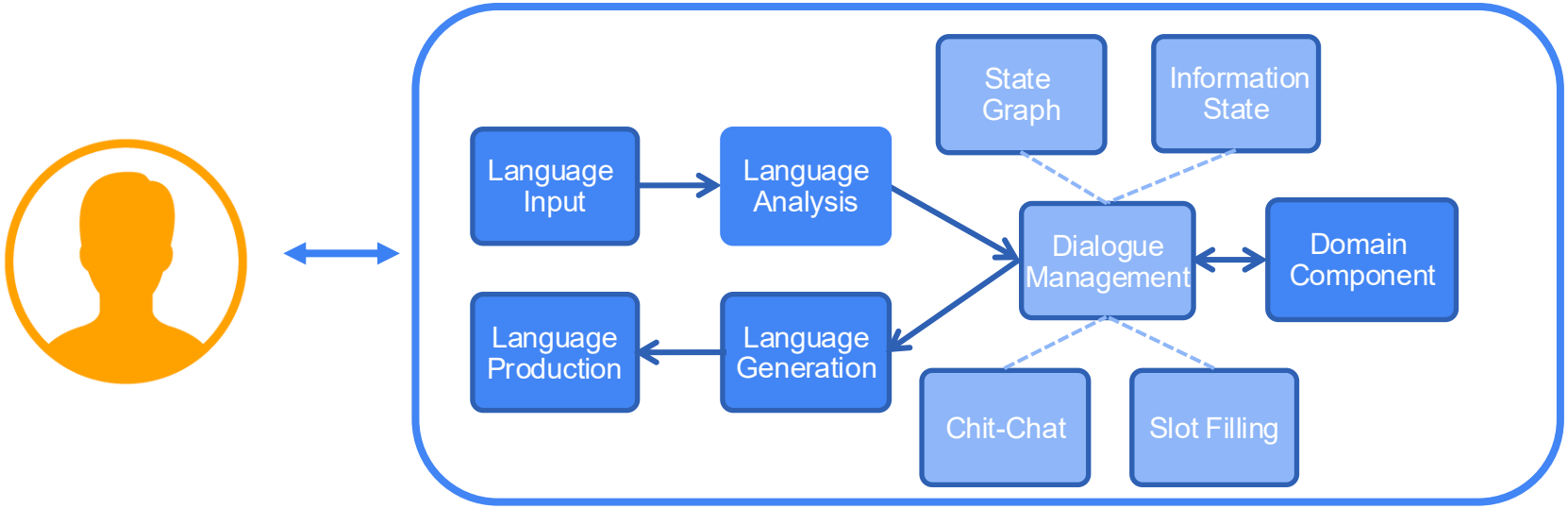
- These two elements are essentially universal. How they are implemented in practice can vary a lot between different dialogue managers.

# Dialogue Manager Types



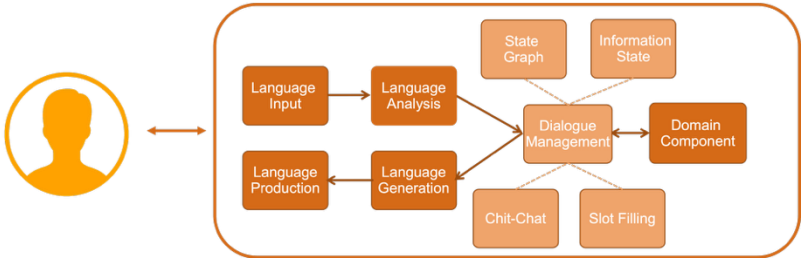

Well Defined Paradigms

# Dialogue Management



| User | P | System |
|---|---|---|
| Hello | 1.0 | Hi, I'm pleased to meet you |
| What is your name? | 1.0 | My name is Stupido |
| I feel <emotion> | 1.0 | Oh, why do you think you feel <emotion> |
| Goodbye | 0.5 | Bye bye |
| Goodbye | 0.5 | Come back soon |
| …. | …. | …. |
| …. | …. | …. |

*Chit-Chat Dialogue Management*



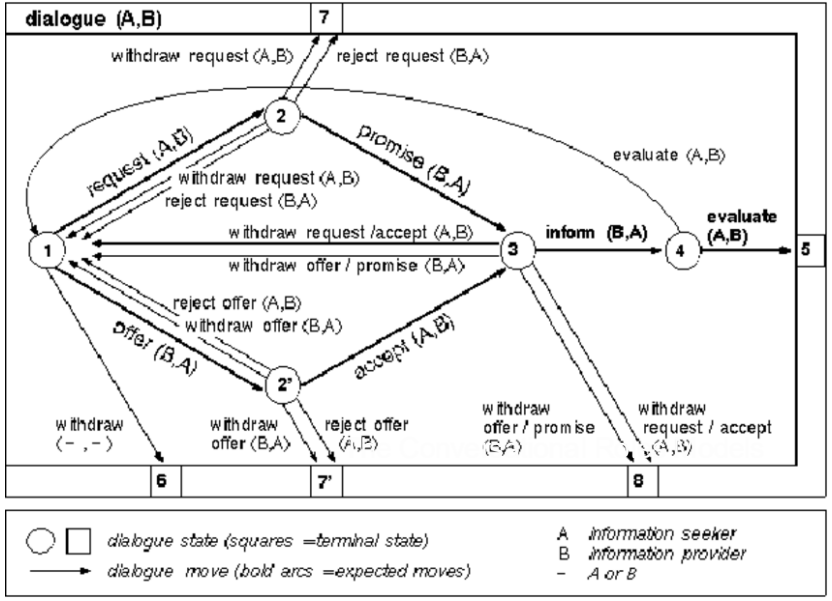

This is often the easiest type of conversational system to build and is based around the concept of initiate-response pairs in language. In their basic form, there is very little context understanding, but a simple FAQ bot would be.a modern version of this.

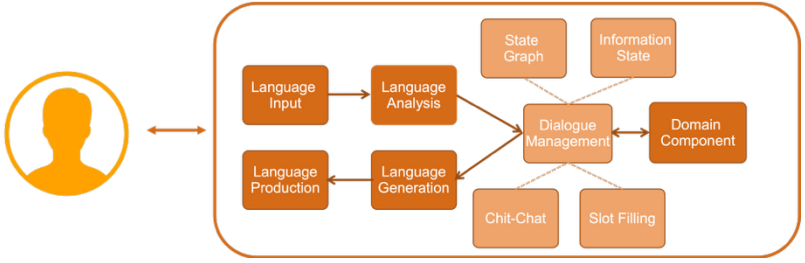
# Pairs in Interaction

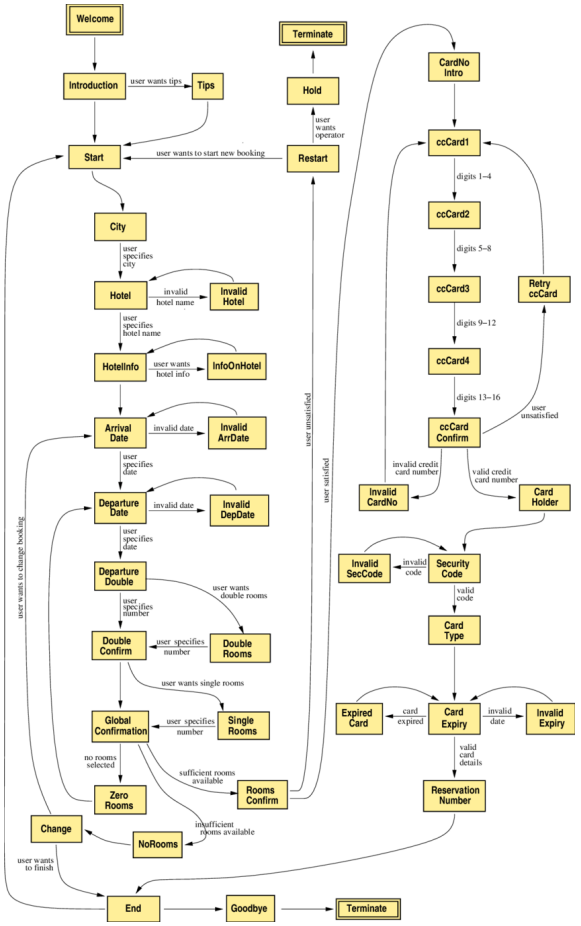
| Initiation | Response |
| --- | --- |
| Hi | Hello |
| What time is it? | 2:01 |
| Play The Beatles | Sure |
| I'm bored | Poor you |
| Question | Answer |
| Inform | Acknowledgement |
| Instruct | Deny |
| Greet | Greet |

Regardless of whether we characterize these as turns, acts or something else, the pairing of an initiating utterance with a response is a particularly important concept in conversation

# State Graph Dialogue Management



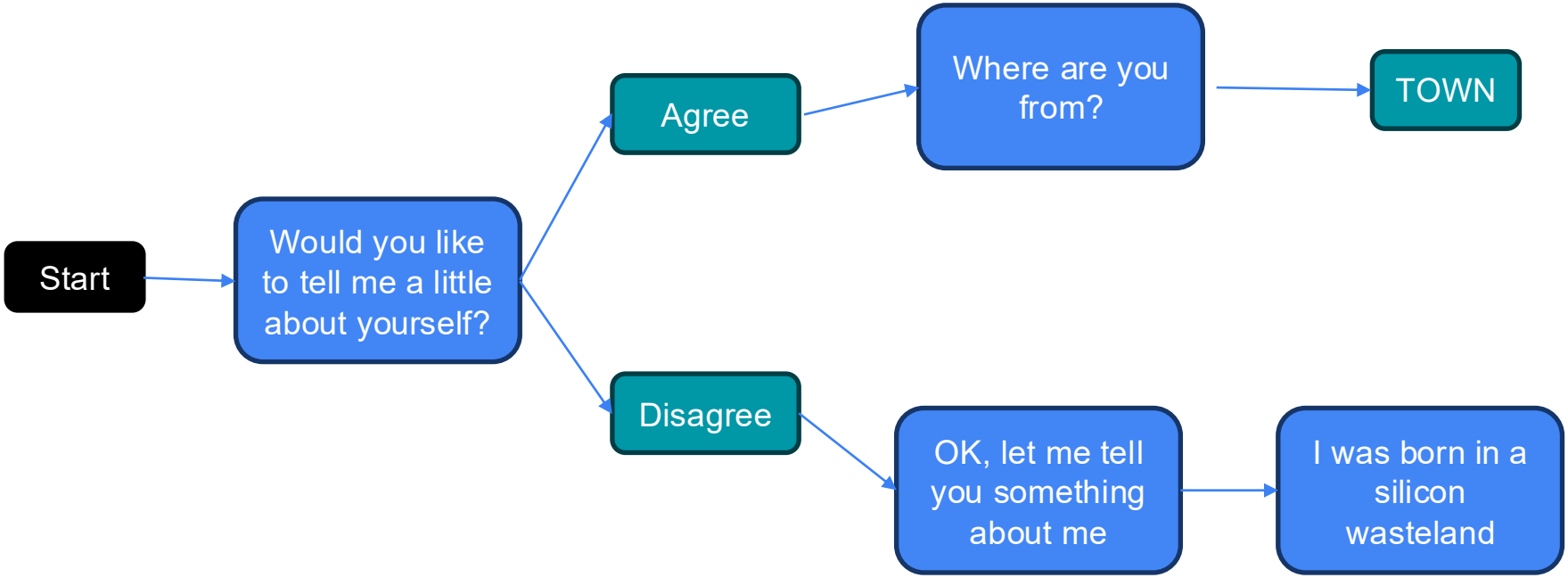



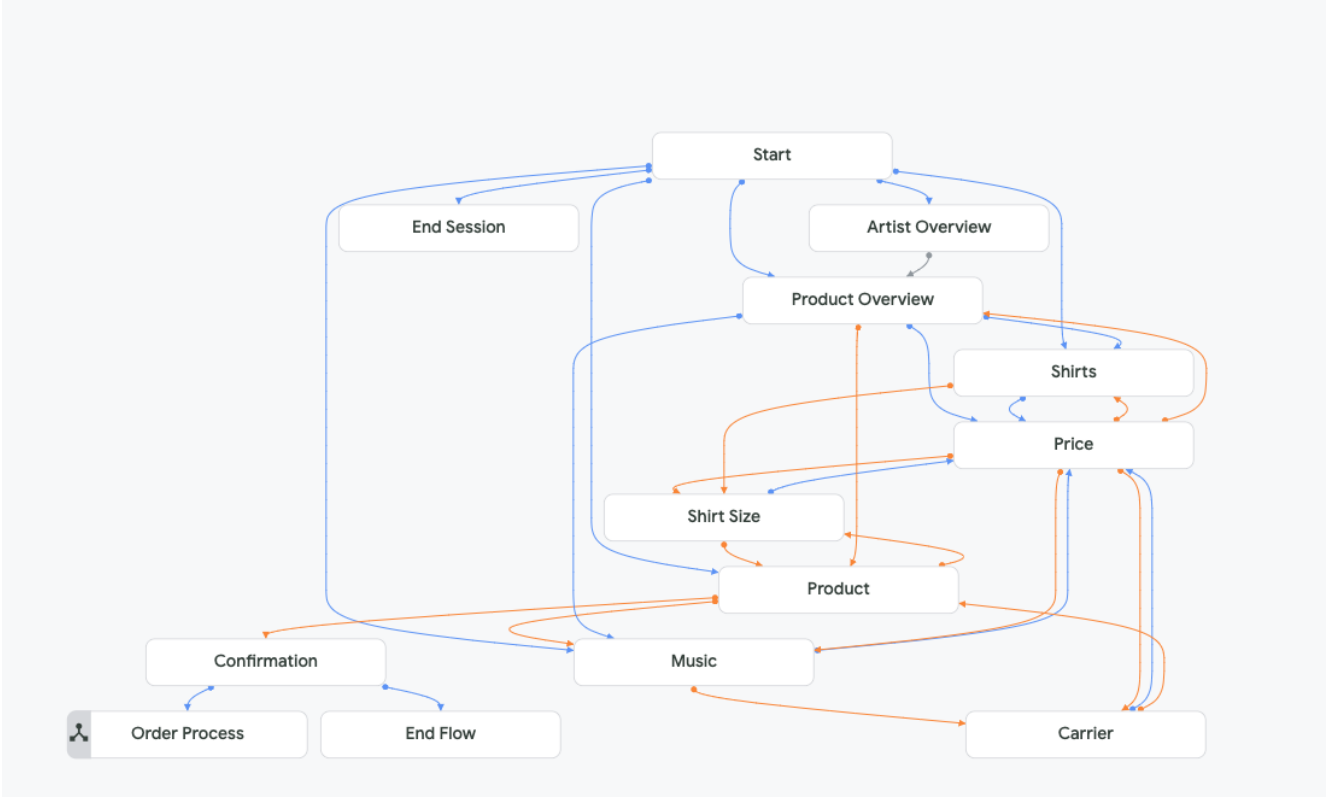
# State / Graph Based Dialogue Management

- We can model dialogue as a graph structure that we transition through. This is very common in commercial systems and relates back to earlier systems such as Interactive Voice Response (IVR) systems for phones
- While technically nodes in the graph should be states and connections should usually be actions that cause us to transition to new states, in practice the nodes are usually ‘actions’ performed

# Simplified Example



# Google DialogueFlow Example of Complex Flow



(RULE 3.1) RULE: **integrateUsrAsk**

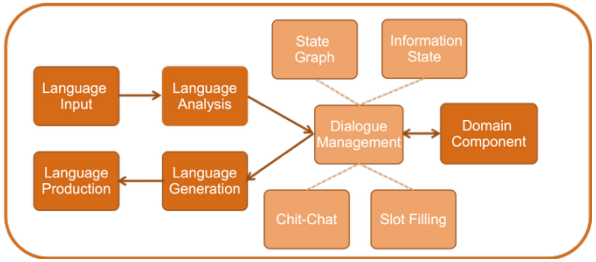
CLASS: integrate

PRE:
- $/SHARED/LU/SPEAKER==usr
- fst($/PRIVATE/NIM, ask($Q$))
- $SCORE=$Score$
- $Score > 0.7$
- $DOMAIN :: plan($Q$, $_Plan$)

EFF:
1. pop(/PRIVATE/NIM)
2. push(/PRIVATE/AGENDA, icm:acc*pos)
3. add(/SHARED/LU/MOVES, ask($Q$))
4. if_do($Score \leq 0.9$, push(/PRIVATE/AGENDA, icm:und*pos:usr*issue($Q$)))
5. if_do(in($/SHARED/QUD, $Q$) and not fst($/SHARED/QUD, $Q$), push(/PRIVATE/AGENDA, icm:reraise:$Q$))
6. push(/SHARED/QUD, $Q$)
7. push(/PRIVATE/AGENDA, respond($Q$))

*Information State Dialogue Management*





# Rule Based Dialogue Management

- Focusing on the ‘knowledge’ or information gathered.
- This is the ‘old-AI’ approach, very much rule-based
  - Nice in theory
  - In practice ‘hand-crafting’ these rules is much too complicated.
- Ironically the ChatGPT type solutions provide the best example of what these earlier technologies were trying to achieve
- But in the mean-time something more middle-ground is needed

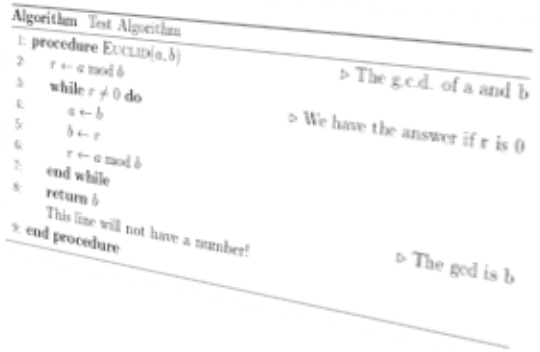
# Slot Based Dialogue Management

- Often the practical middle ground
- Separate the information needed from specific tasks from the process used to gather it
- Focus on filling in slots within a form
  - Slots are defined in terms of
    - The type of information required, i.e., location
    - Optionally the role the information plays in the form, i.e., destination
    - The language used to prompt for the content
    - The language that a user might be expected to use to fill the slot
  - The last two elements above could be generated automatically in theory for a range of slots

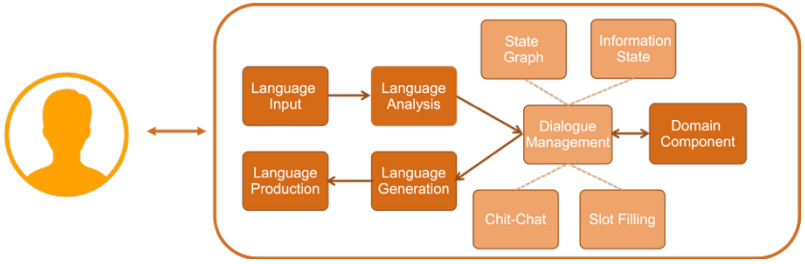
| FoodType |
| --- |
| Burgers |
| Chinese |
| Korean |
| Sushi |
| …. |

| Food | Day | Time | Location | Cost |
| --- | --- | --- | --- | --- |
| Chinese | | | Dublin 7 | cheap |





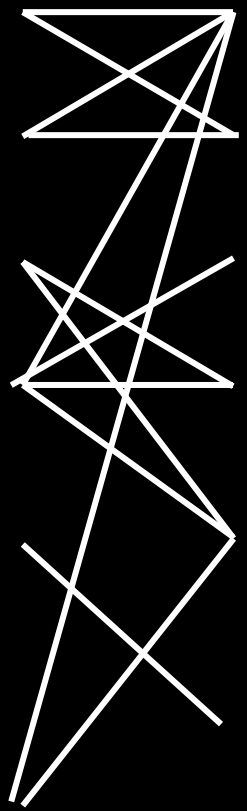

*Slot Based Dialogue Management*



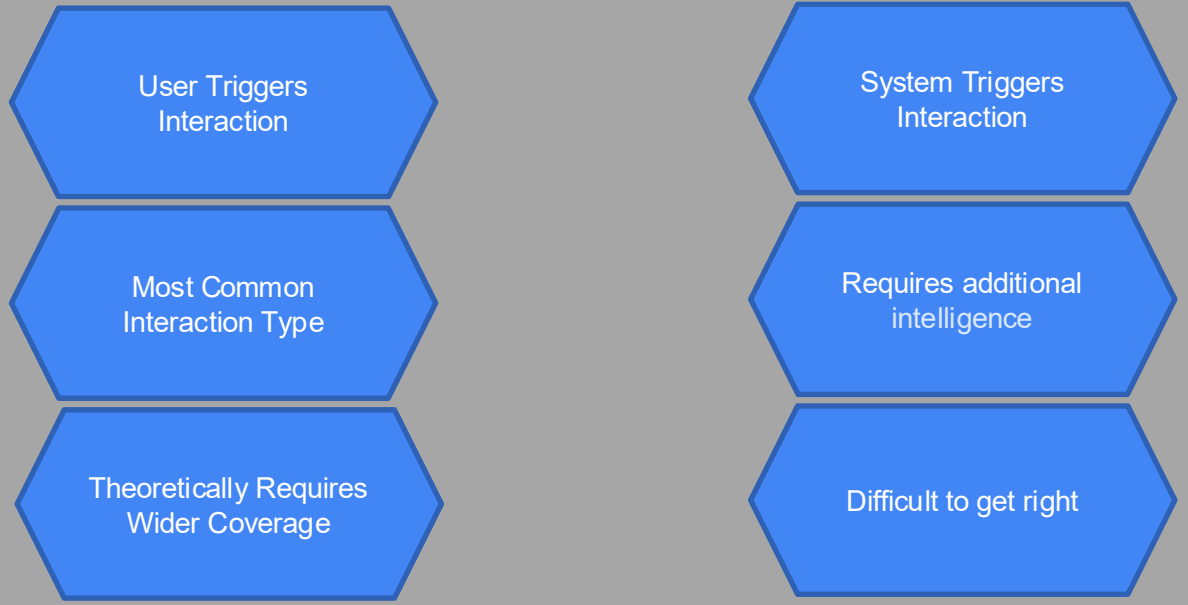
# Data Structures vs Algorithms



| Data Structures | Algorithms |
| --- | --- |
| Raw Text | Hand Crafted Rules |
| State Graphs | State Transitions |
| Knowledge Graphs | Algorithmic Solutions |
| Slots & Frames | State Estimators |
| Distributed Representations | Non-Deterministic Action Planners |
| Databases | Seq-to-Seq and Transformers |
| Pseudo-Logical | |



# User Initiative Vs System Initiative



# Metrics for Comparison

Age

Understandability

Scalability

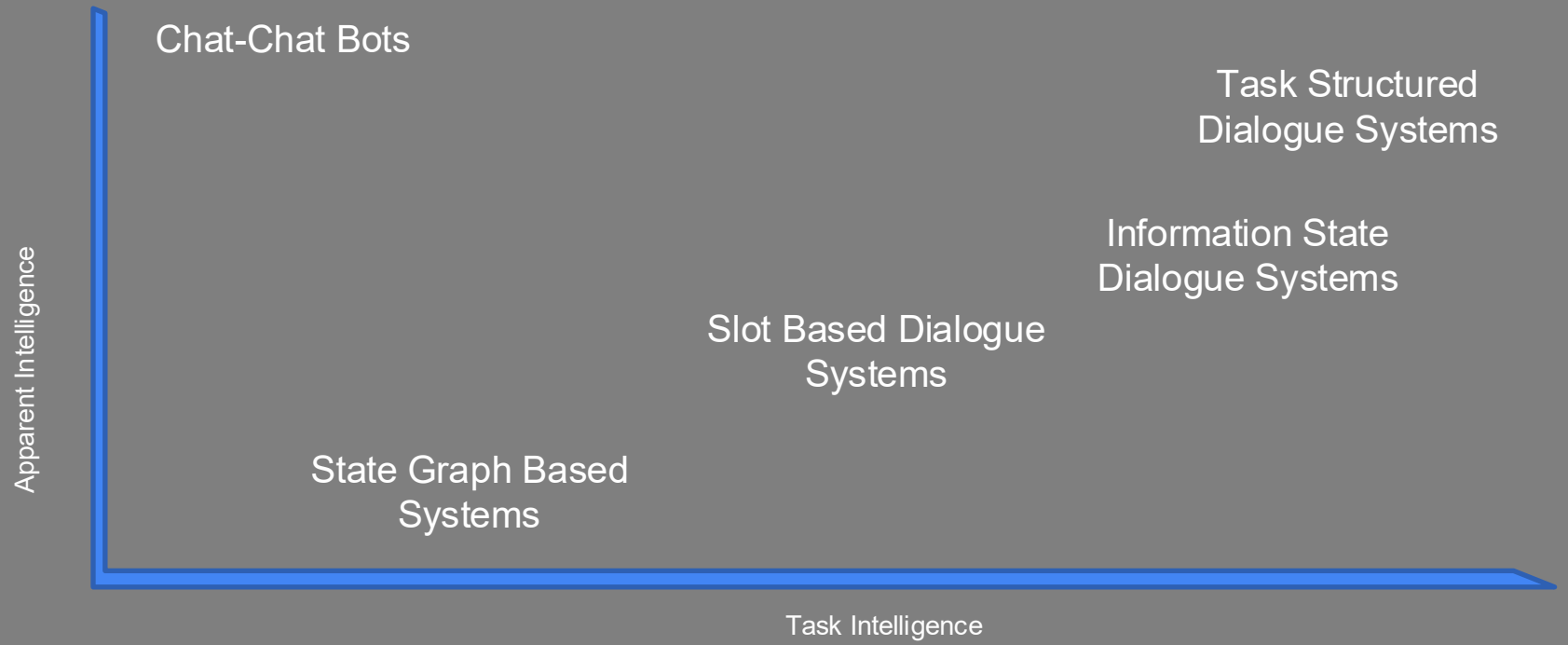
Enterprise Level Tool Support

Long-Term Potential

Apparent Intelligence

Task Intelligence

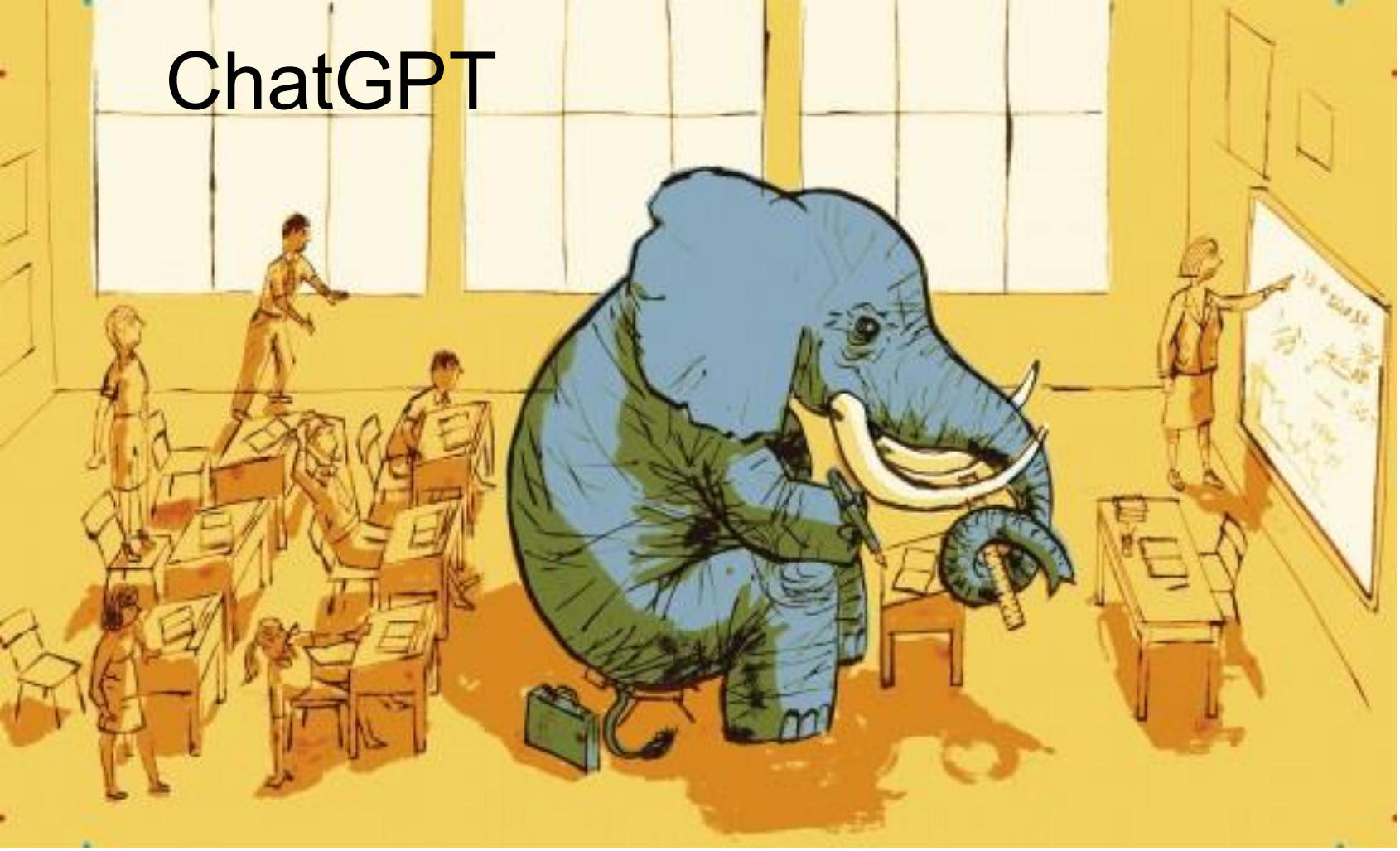
# Conventional Types by Intelligence



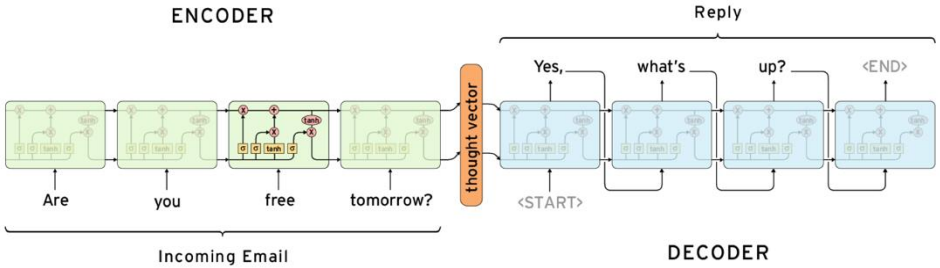
# DM Technology Options



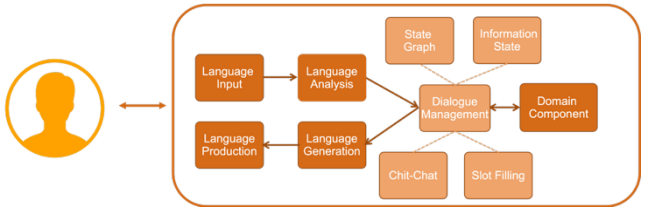
# ChatGPT



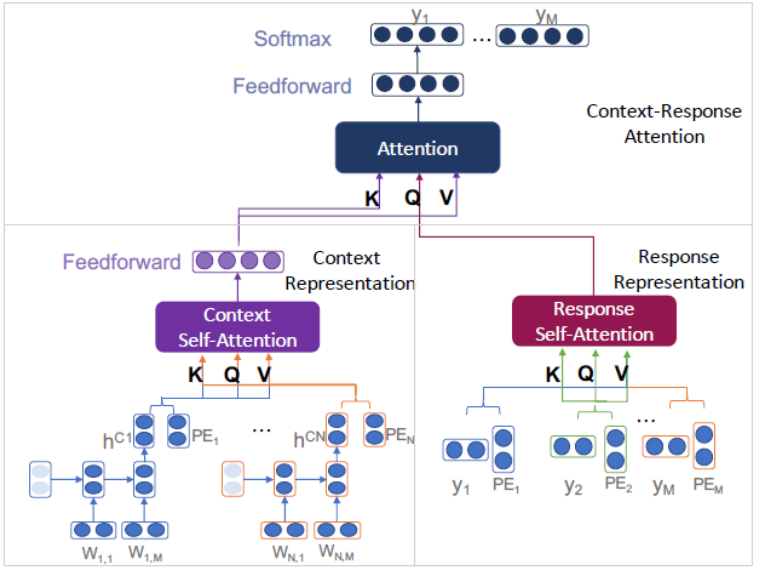
# *Sequence to Sequence Dialogue Management*

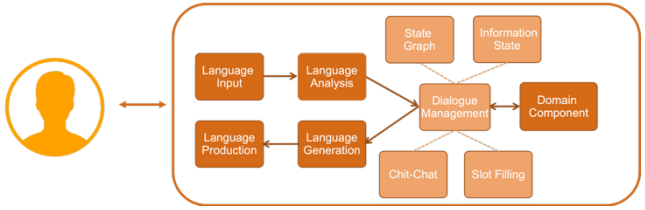



Source https://research.googleblog.com/2015/11/computer-respond-to-this-email.html

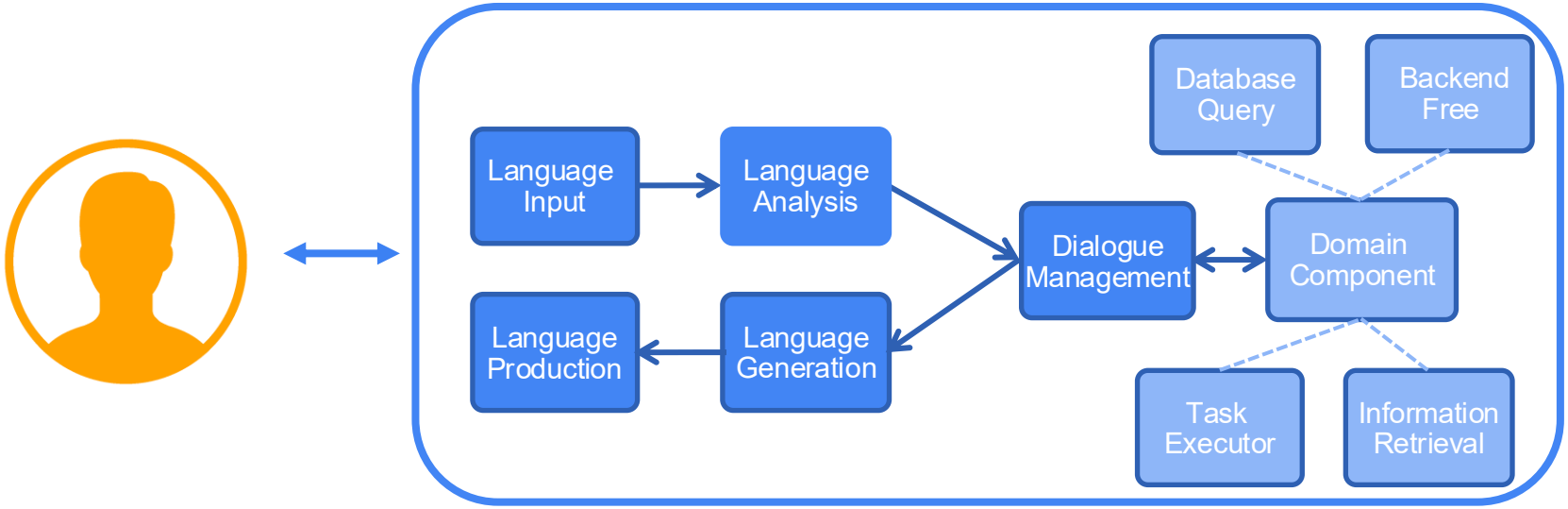


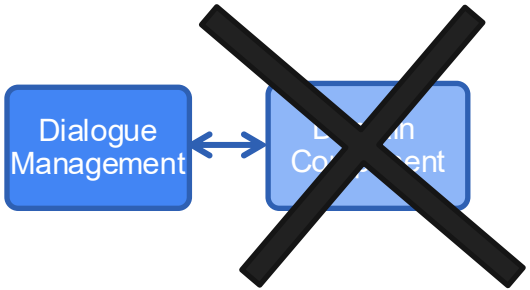
# *Sequence to Sequence Dialogue Management*



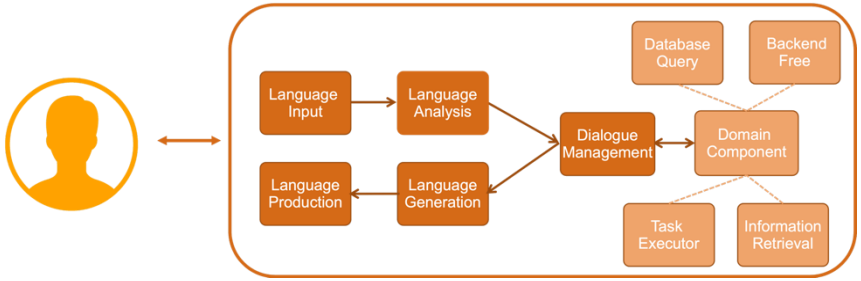



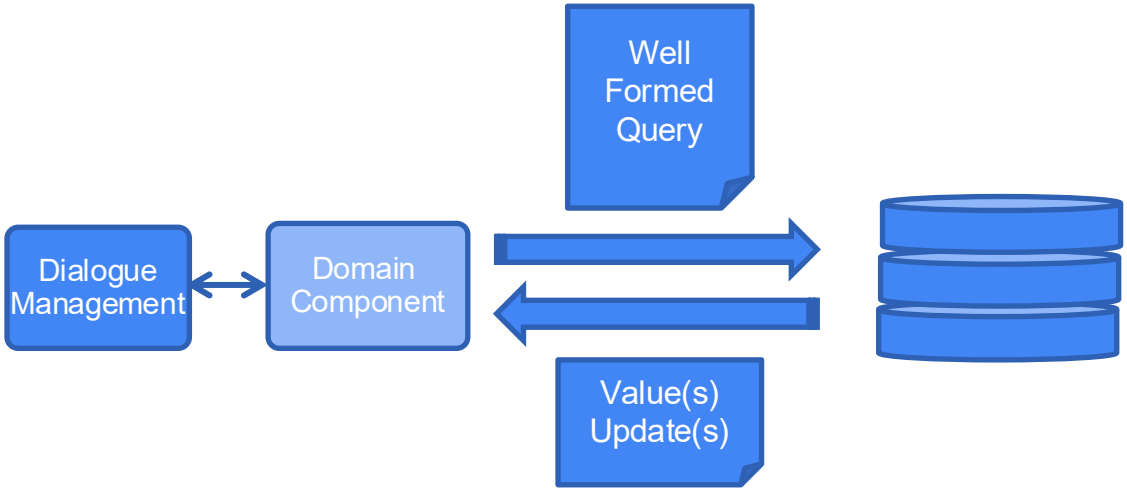
# Domain Components





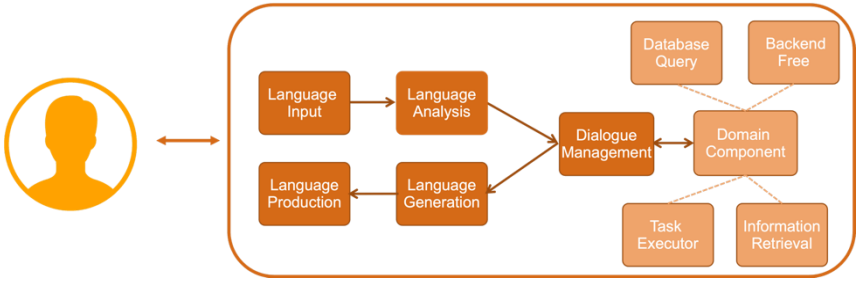

*Backend Free*

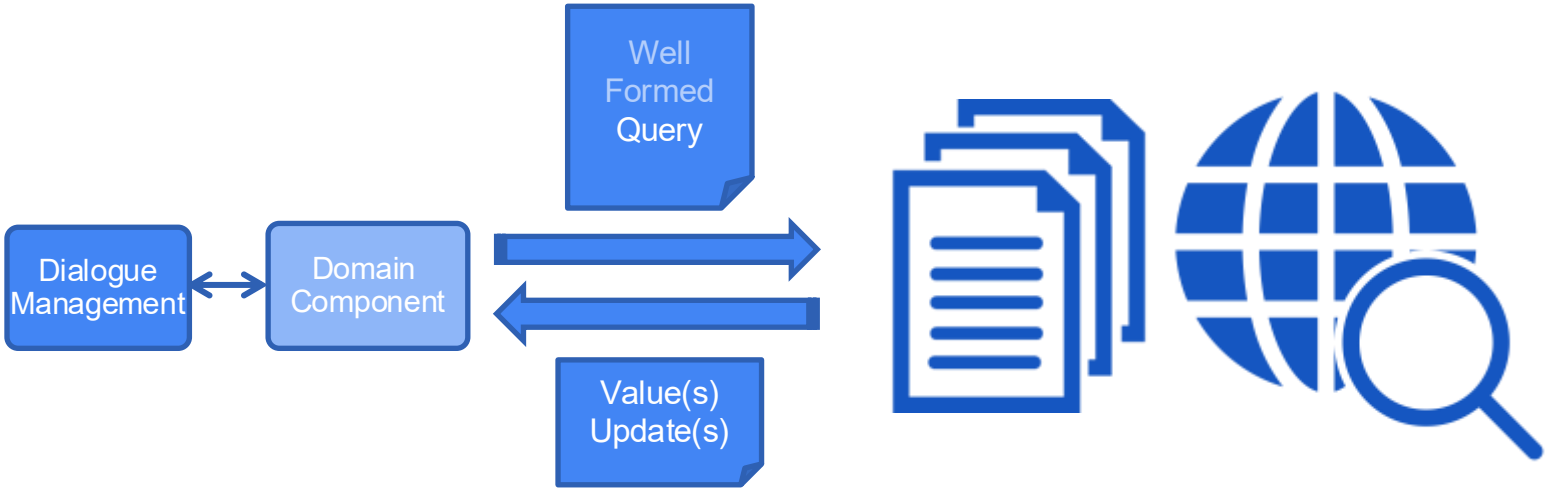




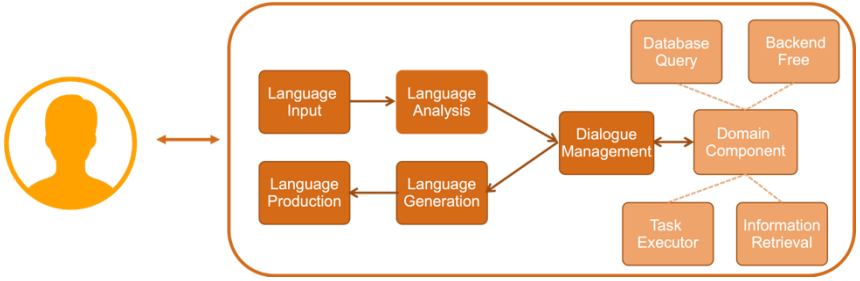

| Name | Phone |
|---|---|
| Adam Smith | (101) 345-678 ext 90 |
| Krissy Jhon | (101) 244-556 ext 78 |
| Scarlett Beck | (801) 945-675 ext 90 |
| Monica Will | (705) 543-678 ext 90 |
| Dulcy Gib | (111) 789-034 ext 56 |
| Andrew Ju | (101) 213-456 ext 70 |

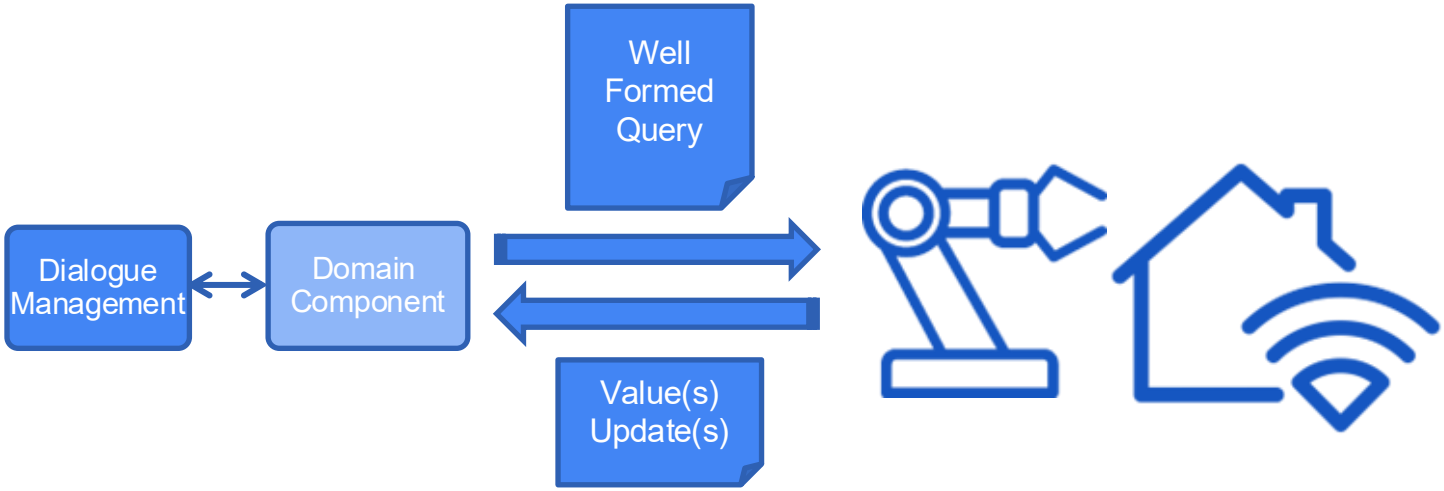
*Database Query*



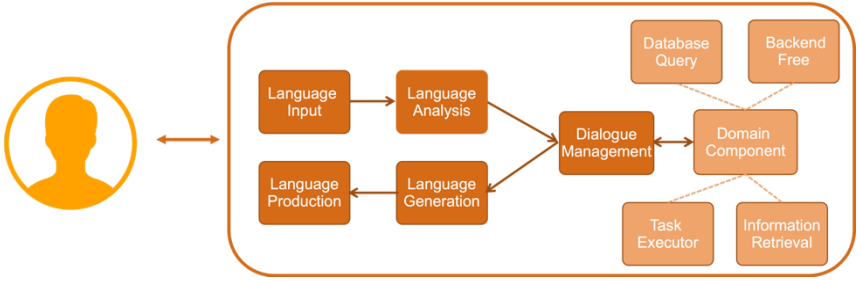



*Information Retrieval*






*Task Executor*



# Lab Work

# Lab Work

- Work your way through the lab material
- Make sure you understand what is going on with each example
- Load a Wikipedia page on the topic of your choice from a text file.
    - Investigate tools to retrieve a page from the web, save the text and load it.